

Formation Engine Methodology

Section 1 — Foundational Thesis and Scope

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1 Motivation: The Emergence Boundary

Coarse-grained models often assume structural statistics rather than deriving them from underlying interaction physics.

This creates a practical discontinuity: we can accurately describe systems *once structure is known*, but we lack reproducible tools for generating physically admissible structural ensembles at intermediate scales. The result is an implicit reliance on heuristic structure selection and opaque initialization pipelines.

The formation engine described in this work is motivated by this specific gap. It is not intended to replace higher-fidelity physics, nor to predict materials in isolation. Instead, it provides a reproducible computational instrument for generating and evaluating candidate structures under explicit physical assumptions, allowing structure to be treated as a measurable, auditable simulation output rather than an implicit starting condition.

1.1 Where Systems Stop Being Molecules

The transition from molecular to material behavior does not occur at a specific size threshold. Rather, it emerges when:

- Collective interactions dominate local bonding energetics
- Long-range order becomes thermodynamically significant
- Surface-to-volume effects no longer control system properties
- Statistical ensembles replace individual configuration determinism

For a 10-atom water cluster, the system is fundamentally molecular: each molecule retains discrete hydrogen-bonding identity, the cluster has a well-defined lowest-energy structure, and thermodynamic behavior is dominated by entropic mixing of conformers.

For bulk liquid water at 300 K with 10^{23} molecules, the system is fundamentally statistical: individual molecules are indistinguishable, the macroscopic state is defined by temperature and density, and thermodynamic behavior emerges from ensemble averages.

The methodological challenge arises in the intermediate regime: 100-atom clusters, 1000-atom nanoparticles, 10,000-atom crystalline seeds. These systems are:

- Too large for brute-force quantum methods

- Too small for continuum approximations
- Too structured for pure statistical mechanics
- Too flexible for rigid lattice models

Existing tools tend to project these systems into one of the two established regimes — either treating them as large molecules (with increasingly intractable quantum calculations) or as small bulk samples (with questionable validity of thermodynamic limits). The formation engine is designed to operate natively in this intermediate regime.

1.2 The Opacity of Initialization

A recurring pattern in simulation workflows is the treatment of structure as an exogenous input. A typical protocol might read:

“The system was initialized in a face-centered cubic lattice with lattice parameter 5.64 Å and equilibrated for 100 ps at 300 K before production sampling.”

This statement embeds several assumptions:

1. The initial lattice choice (FCC) is physically reasonable
2. The equilibration time is sufficient to erase initial condition artifacts
3. The final state is independent of the initialization path
4. The structure has actually reached a relevant local minimum

For well-studied systems (e.g., argon), these assumptions are often justified by decades of experimental validation. For novel compositions or non-equilibrium conditions, they represent unverified choices.

The problem is not that these choices are wrong — often they are correct — but that they are *opaque*. The physical reasoning behind the initialization is not recorded, the sensitivity to alternative initializations is not tested, and the provenance of the starting structure is not preserved.

This creates a reproducibility gap. If someone attempts to replicate the simulation with a different initialization protocol, differences in results cannot be cleanly attributed to:

- Physical sensitivity to initial conditions
- Implementation differences in the integration scheme
- Numerical precision artifacts
- Errors in the reported protocol

The formation engine addresses this by treating initialization not as a preparatory step but as a **primary experimental parameter**. Every structure generated by the engine carries full provenance: the seed that controlled stochastic sampling, the temperature and density conditions, the force field parameters, and the convergence criteria. This transforms initialization from an opaque procedure into an auditable record.

1.3 The Missing Middle Layer

The computational materials science landscape has clear strengths at two extremes:

Small-system quantum methods deliver chemical accuracy for molecules and clusters up to ~ 100 atoms. Density functional theory (DFT), coupled cluster methods, and configuration interaction provide energies and forces that can be directly compared to spectroscopic measurements. The limitation is computational cost: scaling ranges from $\mathcal{O}(N^3)$ for DFT to $\mathcal{O}(N^7)$ for high-level post-Hartree-Fock methods.

Bulk statistical models describe materials at thermodynamic equilibrium using ensemble-averaged properties. Coarse-grained models, continuum mechanics, and phenomenological theories provide tractable descriptions of systems with 10^{20} particles. The limitation is structural detail: these models assume structure rather than generating it.

Between these regimes lies a middle layer — systems with 100 to 10,000 atoms — where neither extreme is fully satisfactory. This is precisely the scale where:

- Nanomaterials exhibit size-dependent properties
- Crystallization nuclei first exhibit bulk-like order
- Self-assembled structures reach functional complexity
- Surface reconstruction competes with bulk stability

Current approaches to this middle layer typically involve:

1. **Empirical structure databases:** Start from known crystal structures (ICSD, COD) and perturb them. This works only for compositions with documented structures.
2. **Evolutionary algorithms:** Propose structures randomly and select based on energy. This often converges slowly and produces unphysical intermediate states.
3. **Machine learning surrogates:** Train on existing data to predict properties. This requires training data and does not generalize beyond the training distribution.
4. **Heuristic builders:** Use chemical intuition (VSEPR, Pauling rules) to construct starting points. This relies on chemist expertise and may miss non-intuitive structures.

Each approach has merit. The formation engine does not replace them. Instead, it provides an alternative pathway: generate structures through physically motivated dynamics, classify the results, and build ensembles that can be interrogated statistically.

The key distinction is that the formation engine generates structure ensembles *under declared physical constraints*, not through black-box optimization or data-driven fitting. This makes it complementary to, rather than competitive with, existing tools.

2 The Failure Mode of Current Toolchains

To understand the design philosophy of the formation engine, it is useful to examine where existing simulation toolchains encounter limitations. This is not a critique of individual methods — each is successful within its intended domain — but an observation about the *integration gaps* between them.

2.1 Quantum Methods: The Tractability Ceiling

Quantum chemistry provides the gold standard for molecular energetics and electronic structure. However, practical quantum calculations face hard computational limits:

- **DFT**: $\mathcal{O}(N^3)$ scaling limits routine calculations to ~ 200 atoms
- **MP2**: $\mathcal{O}(N^5)$ scaling limits routine calculations to ~ 50 atoms
- **CCSD(T)**: $\mathcal{O}(N^7)$ scaling limits routine calculations to ~ 20 atoms

Linear-scaling and fragment-based methods extend these limits, but they introduce approximations that must be validated case-by-case.

For formation problems, quantum methods face a more fundamental challenge: **they are not inherently generative**. A quantum calculation evaluates the energy and forces at a single geometry. To explore the configuration space of a 100-atom cluster:

1. Generate candidate structures (via some other method)
2. Run quantum calculations on each candidate
3. Compare energies to identify low-energy structures

This is computationally tractable only if the number of candidates is small (dozens, not thousands). Exhaustive sampling is impossible.

The formation engine does not replace quantum methods. Instead, it generates physically plausible candidate structures that can then be refined with higher-fidelity calculations. This is the standard protocol in computational chemistry: coarse search followed by fine refinement.

The distinction is that the formation engine makes the coarse search reproducible and auditable.

2.2 Force Field Molecular Dynamics: The Generative Gap

Classical molecular dynamics (MD) using empirical force fields is the workhorse of biomolecular simulation and materials modeling. It scales to millions of atoms and microsecond timescales.

However, force field MD is rarely used as a *structure discovery* tool. The typical workflow assumes the starting structure:

1. Start from experimental structure (PDB, CIF)
2. Equilibrate to the target temperature
3. Sample from the equilibrated ensemble

This works beautifully for studying dynamics around a known structure — protein folding, ligand binding, phase transitions — but it does not generate structure de novo.

Attempts to use MD for structure discovery encounter three challenges:

Initialization bias If the initial configuration is far from equilibrium (e.g., random placement of atoms), the system often gets trapped in high-energy metastable states rather than finding the global minimum.

Timescale separation Structural relaxation may require milliseconds, while MD timesteps are femtoseconds. The resulting 10^9 -step simulations are often impractical.

Ergodicity breaking At low temperatures, the system may fail to cross energy barriers between basins, meaning the sampled ensemble depends on the starting configuration.

Enhanced sampling methods (replica exchange, metadynamics, umbrella sampling) address these issues but require careful tuning and are not fully automated.

The formation engine uses MD not as an equilibration tool but as an **exploration tool**. Short dynamical runs at elevated temperature allow the system to cross barriers. Periodic quenching extracts local minima. Multiple seeds generate an ensemble of structures. The goal is not to sample thermal fluctuations but to discover stable configurations.

2.3 Coarse-Grained Models: The Structure Assumption

Coarse-grained and continuum models describe materials at scales where atomistic detail is impractical. These models assume:

- Local structure is known (e.g., amorphous, crystalline FCC)
- Interactions are parameterized phenomenologically
- Thermodynamic properties emerge from ensemble statistics

This works when the material structure is well-characterized. For novel compositions or non-equilibrium conditions, the structure assumption becomes a liability.

Consider modeling a binary alloy A_xB_{1-x} . A continuum model might assume:

- Phase-separated domains with sharp interfaces
- Random solid solution with uniform mixing
- Ordered superlattice with specific stoichiometry

Each assumption leads to different predicted properties. Without experimental guidance or atomistic calculations, choosing the correct structure is guesswork.

The formation engine provides a bridge: generate atomistic structures at intermediate scales, extract structural statistics (coordination numbers, radial distribution functions), and use those statistics to parameterize coarse-grained models.

This is not a replacement for coarse-grained modeling but a way to ground coarse-grained assumptions in atomistically generated structure.

2.4 Data-Driven Methods: Correlation Without Constraint

Machine learning has transformed materials informatics, enabling rapid screening of millions of hypothetical compounds. However, ML models face a fundamental limitation: they interpolate within the training distribution but extrapolate poorly outside it.

For structure prediction, this means:

- **Training phase:** Learn patterns from known crystal structures (ICSD, Materials Project)

- **Prediction phase:** Propose structures for new compositions by interpolating learned patterns

This works when the new composition is chemically similar to the training set. For truly novel compositions — unusual stoichiometries, high-pressure phases, metastable polymorphs — the model has no physical constraint to guide it.

Generative models (VAEs, GANs, diffusion models) address this by learning the *distribution* of structures rather than individual mappings. However, they still face the same limitation: the learned distribution reflects the training data.

The formation engine provides an alternative: generate structures through physical dynamics rather than statistical patterns. This does not eliminate the need for ML (ML can still be used for post-processing and classification) but ensures that the generated structures satisfy basic physical constraints (energy conservation, force balance, thermodynamic consistency).

In this framework, ML becomes a *filter* on physically generated candidates rather than the generator itself.

3 Philosophy: Physics-First Generative Simulation

The formation engine is built on four core principles that distinguish it from typical simulation frameworks.

3.1 Axiom 1: Explicit Units Everywhere

Every quantity in the framework has explicit units, and unit conversions are performed through named constants with full derivations.

The Boltzmann constant:

$$k_B = 0.001987204 \text{ kcal}/(\text{mol}\cdot\text{K})$$

This is not $k_B = 1$ in “simulation units.” This is the exact value in the framework’s declared unit system (Ångströms, femtoseconds, kcal/mol, amu).

The kinetic energy conversion:

$$C_{\text{KE}} = 2390.057$$

This converts $\frac{1}{2}mv^2$ from $\text{amu}\cdot\text{\AA}^2/\text{fs}^2$ to kcal/mol. The derivation is documented:

$$C_{\text{KE}} = \frac{(10^5 \text{ m/s per } \text{\AA}/\text{fs})^2 \times 1.66054 \times 10^{-27} \text{ kg/amu} \times N_A}{4184 \text{ J/kcal}}$$

The Coulomb constant:

$$k_e = 332.0636 \text{ kcal}\cdot\text{\AA}/(\text{mol}\cdot\text{e}^2)$$

Every constant appears in the code exactly as written in the documentation. No hidden normalization. No silent factor-of-two corrections. No “natural units” where $\hbar = c = k_B = 1$.

Why this matters: Unit errors are the most common source of catastrophic silent failures in simulation codes. A missing Avogadro constant, an incorrect unit conversion, a swapped mass/charge ratio — these produce results that look reasonable numerically but are physically meaningless.

By making units explicit, the framework ensures that dimensional analysis can catch errors at compile time (via strong typing) or runtime (via unit-aware assertions).

3.2 Axiom 2: No Hidden Normalization

Many simulation codes internally normalize quantities to simplify numerics:

- Distances scaled by the van der Waals radius: $r \rightarrow r/\sigma$
- Energies scaled by the well depth: $U \rightarrow U/\varepsilon$
- Temperatures scaled by the critical temperature: $T \rightarrow T/T_c$

These “reduced units” make sense for single-component Lennard-Jones systems where every particle has the same σ and ε . They become a source of confusion and error in multi-component systems where different atom types have different normalization factors.

The formation engine never normalizes. Positions are always in Ångströms. Energies are always in kcal/mol. Temperatures are always in Kelvin.

If you read a position from the state and write it to a file, the number you see is the position in Ångströms. No hidden factors. No back-conversion. No ambiguity.

3.3 Axiom 3: No Silent Domain Switching

Some frameworks automatically switch between models based on system properties:

- Use quantum mechanics for small systems ($N < 50$)
- Use force fields for medium systems ($50 < N < 10,000$)
- Use coarse-grained models for large systems ($N > 10,000$)

This is a reasonable design for automated workflows, but it introduces a reproducibility hazard: the model selection logic is implementation-dependent and may not be documented.

The formation engine requires explicit model selection. The user specifies:

- The force field (Lennard-Jones + Coulomb, or bonded MM, or combined)
- The integrator (velocity Verlet, Langevin, FIRE)
- The boundary conditions (periodic box or open space)

Once specified, the model does not change during the simulation. No adaptive switching. No behind-the-scenes optimization. The physics is declared upfront and applied consistently.

3.4 Axiom 4: Deterministic Core + Stochastic Exploration Layer

The formation engine separates two distinct concepts that are often conflated:

Deterministic physics: Once the system state is known (positions, velocities, identities), the evolution is deterministic. The same initial state always produces the same trajectory. This is Newton’s laws.

Stochastic exploration: The initial state is drawn from a probability distribution. Different random seeds produce different initial states, and therefore different trajectories. This is statistical sampling.

The key insight: **the physics is deterministic; only the sampling is stochastic.**

This matters because:

- Reproducibility is guaranteed: the same seed produces the same result
- Ensemble generation is explicit: different seeds explore different basins
- Convergence can be tested: when the ensemble statistics stabilize, sampling is sufficient

Many frameworks inject stochasticity into the physics itself (e.g., Langevin dynamics with random forces). This is physically justified for modeling thermal fluctuations in contact with a heat bath. But it must be clearly distinguished from stochastic sampling.

In the formation engine:

- NVE dynamics: fully deterministic physics, stochastic initial conditions
- NVT dynamics: stochastic forces for thermalization, but still seeded and reproducible
- FIRE minimization: deterministic once started, but initial velocities are sampled

The rule is simple: **all randomness is controlled by the seed.**

4 The Tool as an Instrument, Not a Claim

This work does not attempt to directly predict new materials, nor does it attempt to replace quantum chemistry or high-fidelity electronic structure methods. Instead, it defines and implements a reproducible structure-generation instrument operating under explicitly declared physical assumptions.

In experimental science, progress rarely begins with discovery; it begins with instrumentation. Microscopes preceded cell theory. Spectroscopy preceded modern quantum chemistry. In this same spirit, the formation engine described here is positioned as an observational and generative instrument for exploring the space between atomic structure and emergent material behavior.

The distinction is intentional. Discovery claims require validation pipelines far beyond the scope of a single computational framework. Instrument claims require only that:

1. The assumptions are explicit
2. The physics is internally consistent
3. The outputs are reproducible
4. The failure modes are observable and classifiable

This work therefore focuses on building a tool that produces physically interpretable ensembles, rather than asserting material novelty or predictive dominance.

4.1 The Microscope Analogy

When Antonie van Leeuwenhoek first observed bacteria under a microscope in 1676, he did not publish “A Theory of Microbial Life.” He published detailed observational reports: drawings of what he saw, descriptions of sample preparation, specifications of lens magnification.

The microscope was not a discovery claim. It was an *instrument that enabled discovery*.

The formation engine follows the same philosophy. It does not claim to predict which crystal structure iron will adopt at 100 GPa. It provides an instrument for:

- Generating candidate structures under specified conditions
- Evaluating their energetic stability
- Classifying their structural motifs
- Recording the full provenance of how they were generated

Other researchers can then take those candidates and refine them with DFT, compare them to experiments, or use them as starting points for enhanced sampling.

The value is not in the final answer but in the *reproducible generation protocol*.

4.2 What Instrument Claims Require

An instrument paper must demonstrate four things:

1. Explicit assumptions Every approximation is documented:

- Classical mechanics (Born-Oppenheimer approximation)
- Pairwise additive potentials (Lennard-Jones + Coulomb)
- Harmonic bonded terms (valid near equilibrium)
- Finite cutoff radius (neglects long-range dispersion tail)

The framework does not hide these. Section 3 of the full methodology document derives every potential, specifies every parameter, and explains where the approximation breaks down.

2. Internal consistency Physical conservation laws are tested:

- Energy drift in NVE dynamics: $|\Delta E|/N < 10^{-3}$ kcal/mol over 10^4 steps
- Momentum conservation: $|\mathbf{P}(t) - \mathbf{P}(0)| < 10^{-10}$
- Temperature stability in NVT: $|\langle T \rangle - T_{\text{target}}|/T_{\text{target}} < 0.05$

These are not performance benchmarks. They are *sanity checks* that the code implements the stated physics correctly.

3. Reproducibility The same inputs produce the same outputs:

- Same formula + same seed + same parameters = bit-identical result
- Cross-platform reproducibility (Linux, Windows, macOS)
- Version-pinned dependencies (library versions recorded in output files)

This is enforced through:

- Seeded random number generation (`std::mt19937`)
- Index-ordered force evaluation (no thread-dependent accumulation)
- IEEE 754 floating-point compliance (no fast-math flags)

4. Observable failure modes When the simulation fails, the failure is classified:

- NUMERICAL: NaN, overflow, loss of precision
- PHYSICS: Atom overlap, unphysical velocities
- OUT-OF-DOMAIN: Undefined parameters, unsupported composition
- CONVERGENCE: FIRE stalled, energy plateau

The self-audit system (Section 11) logs every failure with a minimal reproduction command. This allows systematic debugging rather than case-by-case troubleshooting.

4.3 The Periodic Table as Sole Authority

A central constraint of this work is that the only external data source is the periodic table:

- Atomic number Z
- Standard atomic weight m
- Covalent radii (single, double, triple)
- Van der Waals radius
- Pauling electronegativity
- First ionization energy
- Electron affinity
- Valence electron count

No molecular databases. No empirical crystal structures. No bond-length tables for specific molecules.

The Lennard-Jones parameters (σ , ε) are drawn from the Universal Force Field (UFF) parameterization, which itself was derived from periodic trends. For elements without UFF parameters, carbon-like defaults are used (explicitly flagged in the output).

This constraint serves three purposes:

1. Generality The framework can handle any composition, even ones never synthesized. There is no training set and no out-of-distribution failure.

2. Transparency The data sources are unambiguous. Anyone can verify the values.

3. Testability If the engine predicts the wrong structure for a well-known system, the failure is traceable to:

- Inadequate force field (fixable by parameterization)
- Insufficient sampling (fixable by more seeds)
- Incorrect integration (a bug, fixable by debugging)

There is no hidden dataset to blame.

5 What “Physically Admissible” Means in This Framework

A central concept in this work is **physical admissibility**, which is deliberately narrower than chemical correctness or experimental validation.

A configuration is considered physically admissible if it satisfies:

1. Bounded energy evolution under integration

Energy $E(t)$ remains finite and bounded for $t \in [0, t_{\max}]$. No divergence to ∞ or $-\infty$. No overflow in any energy term.

2. Stable force continuity under applied interaction models

Forces $\mathbf{F}_i$ are continuous in time (no impulses except from explicit collisions). Force magnitudes remain below $10^3 \text{ kcal}/(\text{mol} \cdot \text{\AA})$ (the empirical threshold for Lennard-Jones singularities).

3. Domain-consistent boundary behavior

For periodic systems: positions stay within the primary cell after wrapping. For open systems: the system does not evaporate (kinetic energy does not exceed potential energy by more than a factor of 10).

4. Charge consistency at the system scale

Total charge $Q_{\text{total}} = \sum_i q_i$ remains constant (no spurious charge creation). For neutral systems, $|Q_{\text{total}}| < 10^{-6} e$.

5. Numerical stability under timestep perturbation

Halving the timestep Δt does not change the trajectory qualitatively (energy values may differ by $< 10\%$, but local minima remain local minima).

This definition reflects a practical truth: simulation tools must first ensure internal physical coherence before external chemical realism can be meaningfully evaluated.

5.1 What Admissibility Does Not Imply

Importantly, admissibility does *not* imply:

Thermodynamic equilibrium A configuration can be admissible even if it is metastable. Ethane in the eclipsed conformation is higher in energy than the staggered conformation, but it is still a local minimum and therefore admissible.

Synthetic feasibility A configuration can be admissible even if it cannot be synthesized experimentally. High-pressure phases, strained molecules, and defect-rich structures may all be admissible.

Experimental observability A configuration can be admissible even if it has never been observed. The point of a generative tool is to explore possibilities, not to reproduce known results.

5.2 Why This Definition Matters

The admissibility criterion creates a *first filter* on generated structures. Before comparing to experiments or running expensive quantum calculations, we ask:

Does this configuration survive integration without numerical failure?

If yes, it merits further investigation.

If no, it indicates:

- Force field breakdown (e.g., atoms too close)
- Integration instability (timestep too large)
- Boundary condition error (incorrect periodic wrapping)

This filter is computationally cheap (it runs during generation) and physically motivated (it tests basic physical constraints).

The self-audit system classifies non-admissible configurations and flags them for inspection. This prevents garbage structures from entering the dataset.

6 Deterministic Physics, Probabilistic Exploration

The engine operates on a dual-layer philosophy:

6.1 Deterministic Layer

Classical equations of motion:

$$\mathbf{F}_i = m_i \mathbf{a}_i$$

Given positions $\mathbf{X}(t)$ and velocities $\mathbf{V}(t)$, the forces $\mathbf{F}(t)$ are computed from the potential $U(\mathbf{X})$:

$$\mathbf{F}_i = -\nabla_{\mathbf{x}_i} U(\mathbf{X})$$

The potential is a fixed function (Lennard-Jones, Coulomb, bonded terms). No parameters change during the simulation.

Explicit interaction potentials: Every interaction is defined by closed-form expressions. The Lennard-Jones potential:

$$U_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

The Coulomb potential:

$$U_{\text{Coul}}(r) = \frac{k_e q_i q_j}{r}$$

No black-box force evaluation. No neural network surrogate. The physics is transparent.

Fully specified boundary conditions: For periodic systems: box dimensions (L_x, L_y, L_z) are declared. For open systems: the simulation volume is explicitly unbounded.

No hidden stochastic force injection: In NVE dynamics, there are no random forces. The trajectory is fully deterministic once started.

In NVT dynamics (Langevin), random forces are present:

$$\mathbf{F}_i^{\text{random}} = \sqrt{2\gamma m_i k_B T} \boldsymbol{\xi}_i(t)$$

but these are *explicitly declared* as part of the thermostat model. They are not hidden numerical artifacts.

6.2 Exploration Layer

Stochastic initialization: Initial velocities are drawn from the Maxwell-Boltzmann distribution:

$$v_{i,\alpha} \sim \mathcal{N} \left(0, \sqrt{\frac{k_B T}{m_i}} \right)$$

This sampling is controlled by the random seed. The same seed produces the same initial velocities.

Ensemble generation: Multiple simulations are run with different seeds. Each seed produces a different trajectory. The ensemble of results is analyzed statistically.

Controlled randomization in adjacency proposals: The reaction engine (Section 8) proposes reactions by matching reactive sites. The search over possible pairings is randomized (to avoid exponential enumeration) but seeded.

6.3 Why This Separation Matters

This dual-layer design prevents a common failure mode in modern simulation pipelines: **conflating stochastic sampling with stochastic physics**.

Many frameworks introduce randomness at multiple levels:

- Random initial positions (to avoid bias)
- Random integration noise (to mimic thermal fluctuations)
- Random force cutoffs (to speed up computation)

- Random parameter selection (for hyperparameter tuning)

The result is that reproducibility is lost. Two researchers running the same protocol on the same system may get different results, and it is unclear whether the difference is:

- Physical (different parts of phase space were sampled)
- Numerical (different integration errors accumulated)
- Implementation-dependent (different random number generators)

The formation engine enforces a strict rule:

All randomness is controlled by the seed. The physics is deterministic.

This makes ensemble analysis interpretable. If two seeds produce similar results, it suggests a robust formation. If they produce different results, it suggests competing metastable states. The physics does not have hidden randomness that might obscure this distinction.

7 Reproducibility as a Primary Output

Traditional simulation workflows often treat reproducibility as a post-processing concern:

1. Run simulations
2. Analyze results
3. (Maybe) document parameters in a methods section
4. (Rarely) provide raw data or code

The formation engine inverts this priority. Reproducibility is a *first-class output*, recorded with every simulation artifact.

7.1 The Provenance Record

Each formation carries a structured provenance record:

```
---
structure_state:
  atoms: 18
  formula: H2O (6 molecules)
  hash: a7f3c21b89d4e...
engine:
  version: 0.1.0-dev
  commit: f8a32c1
  build: Release (gcc 11.2.0)
units:
  distance: Angstrom
  energy: kcal_mol
  time: femtosecond
```

```

    temperature: Kelvin
sampling:
  seed: 42
  temperature: 300.0
  density: 0.997
boundary:
  type: periodic
  box: [18.6, 18.6, 18.6]
force_field:
  nonbonded: LJ12-6 + Coulomb
  cutoff: 10.0
  switch: quintic [9.0, 10.0]
scoring:
  priority: 87.3
  stability: converged
  classification: molecular_cluster
validation:
  energy_conservation: 0.00034 kcal/mol/atom
  momentum_drift: 3.2e-11
  temperature_error: 0.017
---
```

This is not optional metadata. It is *part of the output*.

7.2 Cross-Institution Comparison

With full provenance, two independent research groups can:

1. Verify they are running the same protocol (same force field, same integrator)
2. Compare results at the same seed (should be bit-identical)
3. Compare ensemble statistics across different seed ranges (should converge)

If results disagree, the provenance record allows debugging:

- Different engine versions?
- Different compiler optimizations?
- Different random number generator implementations?
- Different floating-point rounding modes?

The goal is to eliminate ambiguity.

7.3 Regression Detection

When the code is updated (new feature, bugfix, refactoring), the regression detector (Section 11.4) reruns a benchmark suite and compares:

- Energy values (should be identical to machine precision)
- Force magnitudes (should be identical to machine precision)
- Final structures (should have RMSD $< 10^{-6}$ Å)
- Classification labels (should be identical)

If any of these change, the update is flagged for review. This prevents silent breakage.

7.4 Long-Term Dataset Integrity

Simulation datasets degrade over time if the generation protocol is not preserved. Published results become irreproducible as:

- Software versions become unavailable
- Documentation is incomplete or ambiguous
- Raw data is lost

The formation engine addresses this by treating every output file as a *self-contained artifact*. The provenance record includes:

- Full parameter specification
- Software version and commit hash
- Complete calculation history (MD steps, minimization convergence)

Given the output file alone, someone can:

1. Check out the exact code version
2. Rerun the simulation with the same parameters
3. Verify that the result matches

This transforms simulation results from transient calculations into persistent experimental records.

8 Explicit Scope Boundaries

To maintain clarity, this work intentionally excludes certain physical phenomena. These exclusions are not limitations but design choices.

8.1 Full Electronic Structure Resolution

The formation engine does not solve the Schrödinger equation. It uses classical potentials parameterized from quantum calculations (UFF, OPLS) or periodic table data.

This means:

- Bond dissociation energies are approximate
- Charge transfer is not captured dynamically
- Spin states and magnetic properties are absent
- Optical spectra cannot be predicted

For systems where these effects dominate (transition metal complexes, radical species, conjugated molecules), the formation engine provides only a coarse description.

However, the classical description is sufficient for:

- Structural motifs (packing, coordination, symmetry)
- Thermodynamic stability ordering (relative energies)
- Mechanical properties (bulk modulus, surface tension)

The strategy is to use the formation engine for coarse screening, then refine promising candidates with DFT or higher-level methods.

8.2 Excited States and Non-Adiabatic Phenomena

The Born-Oppenheimer approximation is implicit: electrons instantaneously adjust to nuclear positions, and the potential energy surface is independent of electronic excitation.

This excludes:

- Photochemistry (light-driven reactions)
- Singlet-triplet crossings
- Conical intersections
- Rydberg states

These phenomena require non-adiabatic dynamics or multi-reference quantum methods, which are beyond the scope of this work.

8.3 Reaction Kinetics Prediction

The reaction engine (Section 8) proposes thermodynamically favorable reactions and estimates activation barriers using linear free-energy relationships. It does *not* predict:

- Rate constants
- Transition state geometries
- Pre-exponential factors
- Tunneling corrections

These require transition-state theory calculations or kinetic Monte Carlo, which are complementary tools.

8.4 Quantum Coherence Modeling

The framework is entirely classical. Quantum effects (zero-point energy, tunneling, delocalization) are not included.

For light atoms at low temperatures (hydrogen below 100 K, helium always), quantum effects can be significant. The formation engine may produce incorrect structures in these regimes.

The domain of validity is therefore: **heavy atoms ($Z \geq 6$) at moderate temperatures ($T > 50$ K).**

8.5 Extension Points for Higher-Fidelity Overlays

These exclusions do not preclude future extensions. The architecture is designed to support:

- Quantum-corrected potentials: $U = U_{\text{classical}} + \Delta U_{\text{QM}}$
- Polarizable force fields (Drude oscillators, fluctuating charges)
- Reactive bond-order potentials (ReaxFF, COMB)

The key requirement is that the classical description remains independently evaluable. The correction is an optional layer, not a replacement.

This ensures that:

- The baseline model is always available
- Failures can be diagnosed at the classical level before invoking higher-fidelity physics
- Computational cost scales gracefully with accuracy requirements

9 Structure as a Primary Simulation Result

The most fundamental shift embodied by this framework is philosophical:

In this framework, structure is not assumed, imported, or heuristically stabilized; it is generated under declared physical rules and recorded as a primary simulation result.

This statement has implications for how simulation data is produced, analyzed, and interpreted.

9.1 Structure is Not an Input Assumption

Traditional workflows:

1. Assume a structure (PDB, CIF, or hand-drawn)
2. Simulate dynamics or properties
3. Validate against experiments

Formation engine workflow:

1. Specify composition, temperature, density

2. Generate structure ensemble through dynamics + quenching
3. Classify and rank structures
4. (Optionally) refine with higher-fidelity methods

The distinction is subtle but consequential. In the traditional workflow, the structure is treated as *given*. In the formation workflow, the structure is treated as *emergent*.

9.2 Implications for Discovery

This shift enables a different style of computational exploration:

Inverse design Instead of asking “What are the properties of structure X?” we can ask “What structures have property Y?”

The formation engine generates candidates. A post-processing filter evaluates the property. Structures that pass the filter are kept.

Metastable polymorphs Many materials have multiple metastable phases (diamond vs. graphite, polymorphs of ice). Traditional simulations often miss these because they start from a single structure.

The formation engine generates ensembles. If multiple distinct structures appear repeatedly across seeds, they represent competing metastable states.

Size-dependent phases Nanoparticles often adopt structures different from the bulk. A 55-atom gold cluster is icosahedral, not FCC.

The formation engine can probe this transition by systematically varying N and observing when the preferred structure changes.

9.3 Structure as Data

By treating structure as an output, the formation engine produces datasets where:

- Structures are labeled by generation conditions (seed, temperature, density)
- Structures carry provenance (software version, force field parameters)
- Structures are reproducible (same inputs $\rightarrow$ same outputs)

These datasets can be analyzed statistically:

- What fraction of seeds produce crystalline vs. amorphous structures?
- How does the energy distribution change with temperature?
- Are there preferred structural motifs (coordination shells, symmetries)?

This is analogous to how high-throughput crystallography produces structure databases. The formation engine produces *computational structure databases* with full provenance.

10 Conclusion: The Instrument Framing

This section has established the foundational philosophy of the formation engine. It is not a discovery tool but an *instrument* for reproducible structure generation.

The key claims are modest:

1. The engine implements classical mechanics under explicit assumptions
2. The outputs are reproducible from declared inputs
3. The failure modes are observable and classifiable
4. The results provide physically admissible candidate structures for further refinement

These are *instrument claims*, not *prediction claims*.

The value proposition is equally modest:

If you need to explore the structure space of a composition under specified thermodynamic conditions, and you want the exploration to be reproducible and auditable, the formation engine provides a way to do that.

This framing is deliberate. It positions the work as:

- Complementary to existing methods (not competing)
- Useful even if predictions are imperfect (because reproducibility has value)
- Testable through self-consistency (not requiring expensive validation)

The subsequent sections (2–13) build on this foundation by specifying:

- The data structures (Section 2)
- The interaction models (Section 3)
- The thermodynamic framework (Section 4)
- The integration schemes (Section 5)
- The formation protocol (Section 6)
- The statistical analysis (Section 7)
- The reaction engine (Section 8)
- The electronic estimates (Section 9)
- The multiscale extensions (Section 10)
- The self-audit system (Section 11)
- The validation doctrine (Section 12)
- The future directions (Section 13)

Each section maintains the same tone: explicit assumptions, transparent limitations, reproducible protocols.

The result is not a revolutionary method but a reliable instrument.

And that is exactly the point.

Transition to §2: With the formation problem defined (what the engine solves and what it does not), Section 2 establishes the fundamental data structures: the State ontology that every operation in the framework reads from and writes to.